

ALI HUSAIN RIZVI

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [GeeksforGeeks](#)

TECHNICAL SKILLS

Programming: Python, C++, SQL

CS Subjects: Data Structures & Algorithms (DSA), Operating Systems, Database Management Systems (DBMS), Computer Networks, Object Oriented Programming (OOP)

Problem-Solving: Algorithm Design, Complexity Analysis, Debugging & Optimization, System Design

Analytical Skills: Logical Reasoning, Pattern Recognition, Data Interpretation

SUMMARY

- Results-driven Software Engineer skilled in designing and developing scalable, high-performance applications. Strong analytical and mathematical aptitude with expertise in Data Structures, Algorithms, and efficient problem solving. Passionate about optimizing system performance, building robust backend services, and creating reliable software that scales.

PROJECTS

SENTINEL — AI Prompt Security Gateway

June 2026

Python, Chrome Extension (Manifest V3), React, Chart.js, FastAPI, JWT

- Built a **Manifest V3 Chrome extension** that intercepts prompts on ChatGPT, Claude, and Gemini, scanning for **API keys, credentials, and PII** before submission with **zero network calls in the detection path**.
- Implemented **5 algorithms from scratch** — Aho-Corasick, Shannon entropy, Trie, Max-Heap, and Merge-Intervals — detecting **24 secret patterns** plus **base64-encoded payloads** via a decode-and-rescan layer.
- Shipped a React + Chart.js dashboard backed by a **FastAPI + JWT** service, 3 tunable sensitivity levels, and **104 tests** covering **ReDoS safety and a false-positive corpus**.

Live Demo: alirizzzv.github.io/SENTINEL | GitHub: github.com/alirizzzv/SENTINEL

ADPULSE — Real-Time Demand-Side Platform

June 2026

Python, LightGBM, Flask-SocketIO, Three.js

- Built a **full-stack Demand-Side Platform (DSP)** that participates in **real-time ad auctions**, prices each bid **in-memory** at **~5 ms**, and sustains a live feed of **7 bids/sec**.
- Trained **2 LightGBM models (CTR + CVR)** on IPinYou competition logs with **12 engineered features**; bid pricing scales up to **11x** the base price with a budget pacer and a hard cap of 300.
- Engineered a **generator-based pipeline** that streams **multi-GB logs** in **O(1) memory**; a Flask-SocketIO + Three.js 3D dashboard renders live KPIs, validated across a **Python 3.9/3.11 CI matrix** with **46 tests**.

Deployed Link: adpulse-457r.onrender.com | GitHub: github.com/alirizzzv/ADPULSE

POSITION OF RESPONSIBILITY

Team Lead — Moksha Fest

Security & Operations | 50,000+ attendee festival, Northeast India

- Led a **12-member team** for **security protocols and risk assessment**, coordinating **real-time incident response** with **zero breaches** across a **3-day event**.

EDUCATION

B.Tech in GeoInformatics

2023 – 2027

Netaji Subhas University of Technology — **7.03 CGPA**

Senior Secondary School (CBSE Class XII)

2022

Bal Bharti Public School, Dwarka — **91%**

Secondary School (CBSE Class X)

2020

Bal Bharti Public School, Dwarka — **81%**

ACHIEVEMENTS / EXTRACURRICULAR

- Solved **300+** Data Structures and Algorithms problems on **LeetCode** and **GeeksforGeeks**.
- Earned a **100% Educational Scholarship** (2018–2022) for sports performance from Class 8 to Class 12.
- Ranked within the **top 10 percentile** in **JEE Main** among **1M+** candidates.
- Achieved **top 1% AIR** in **VITEEE**, qualifying for a top-tier branch seat at VIT University.
- Awarded **3x Gold Medals** at the **CBSE National Championship**, recognized at the national level for athletic excellence.